

DSC2204- Integrative Team Project

The Real Problem

- This is a real requirement from an actual academic institution
- Every semester, hundreds of classes must be scheduled
- Each class needs the right room, tutor, and time slot — all aligned
- One conflict causes a cascade — complaints, last-minute changes, wasted resources
- As supply chain students, you know this problem: it is resource allocation under constraints
- Your job is to understand the problem, structure it, and build something that works

Why This Is Hard

- Multiple resources must align simultaneously — room, tutor, student group, time
- Constraints conflict with each other — solving one can break another
- Hard rules and soft preferences are different problems and need different solutions
- Data is messy — public holidays, odd/even weeks, online vs face-to-face modes
- Scale makes brute force impossible — you need a smarter approach
- Focus on engineering cluster first. There are more than 10 programmes from the cluster (ASE, CVE, DSC, EDE, EPE, ESE, MDME, MEC, METS, NAME, RSE, SBE, EEE, ISE).

The Constraints - Example

- **Hard Constraints (must never be violated)**
 - No two classes in the same room at the same time
 - Tutor cannot teach two classes at the same time
 - Student group cannot attend two classes at the same time
 - Room capacity must be $\geq$ class enrolment
 - Online classes $\rightarrow$ virtual room only; Face-to-face $\rightarrow$ never virtual room
 - Odd/even week courses run on their respective weeks only
 - No classes on public holidays or during term breaks
- **Soft Constraints (avoid where possible)**
 - Avoid online $\leftrightarrow$ face-to-face switch in adjacent slots (same tutor or same group)
 - Avoid tutor idle gaps longer than 2 hours on the same day
 - Avoid student group having more than 4 consecutive teaching hours
 - Avoid only 1-2 hours per day in campus
 - Schedule online classes (synchronous or asynchronous) within one day (Monday or Tuesday) for each programme

Timetabling Project

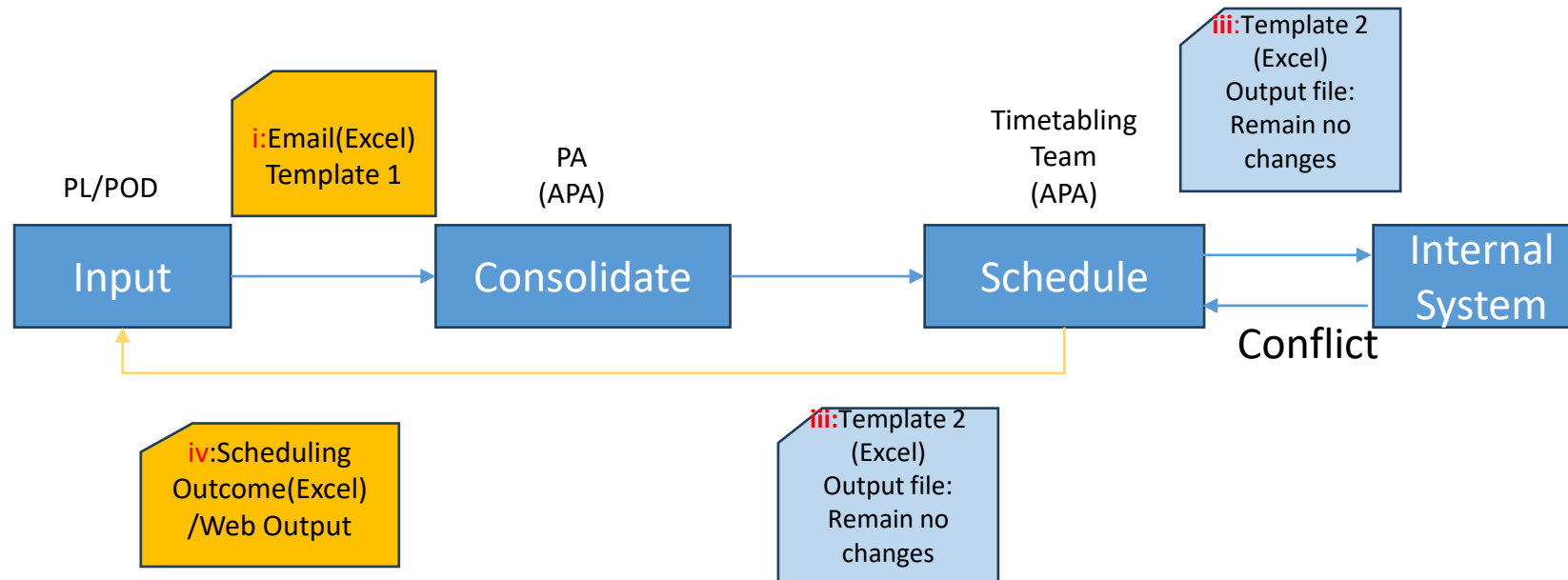
- LLM tools (Claude Code, ChatGPT, etc.) are allowed for coding — but you must understand and explain every decision
- Think like an operations analyst, not a programmer
- You can start by working on the timetable for DSC first, and then extend it to the ENG cluster.
- The timetabling project includes the following segments.

#	Segment	What you build
1	Data Modelling	Structure that represents rooms, tutors, groups, courses, time slots
2	Constraint Engine	Checker that reads a timetable and reports violations
3	Schedule Generator	System that auto-assigns slots satisfying all hard constraints
4	Optimiser	Takes a valid timetable and improves it against soft constraints
5	Interface & Output	Visual display or export tailored to different stakeholders
6	Operation Improvement	Improve the communication and working process

What You Submit

- **Prototype**
 - A working demonstration of the timetabling tool
- **Report**
 - Your understanding of the problem, approach taken, and reflection on what worked and what did not
 - **The process to use AI to get the outcome**
- **Presentation (PPT)**
 - Present your result

Current Process



- Template 1  Microsoft Excel Worksheet
- Template 2  Microsoft Excel Worksheet
- Other info at share drive: [ITP Project_Y2526](#)